Tree-Level $e^-\mu^- \to e^-\mu^-$ Scattering

Caiyi, Jiang

1 The process and the amplitude

Consider the process:

$$e^-(p_1) + \mu^-(p_2) \longrightarrow e^-(p_1') + \mu^-(p_2'), \qquad p_1 + p_2 = p_1' + p_2'. \tag{1.1}$$

According to the momentum conservation, the momentum transfer carried by the virtual photon is

$$q = p_1' - p_1 = p_2 - p_2'. \tag{1.2}$$

We use $g^{\mu\nu} = \mathrm{diag}(1, -1, -1, -1)$, $m_e = 0$, and $m_\mu = m$. The tree-level diagram is the t-channel photon exchange, so $q^2 = t$.

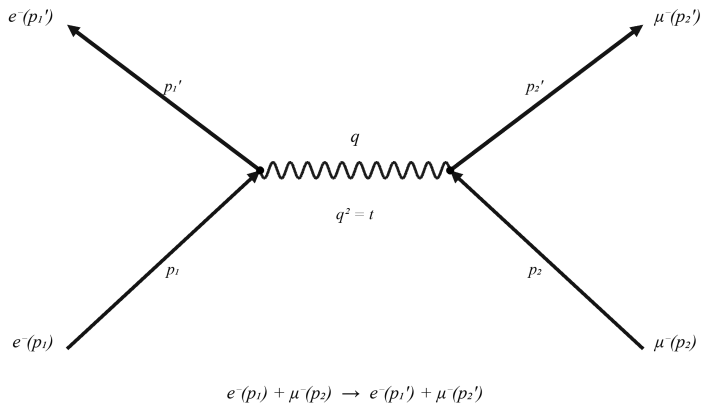


Figure 1: Tree-level t-channel photon exchange for $e^-(p_1) + \mu^-(p_2) \to e^-(p_1') + \mu^-(p_2')$.

According to the QED Feynman rules, for specified external spin labels s_1, s_2, s_1', s_2',

$$i\mathcal{M}_{s_1' s_2'; s_1 s_2} = [\bar{u}_e(p_1', s_1')(-\mathrm{i}e\gamma^\mu)u_e(p_1, s_1)] \frac{-\mathrm{i}g_{\mu\nu}}{q^2} [\bar{u}_\mu(p_2', s_2')(-\mathrm{i}e\gamma^\nu)u_\mu(p_2, s_2)] \tag{1.3}$$

$$= \frac{\mathrm{i}e^2}{q^2} [\bar{u}_e(p_1', s_1')\gamma^\mu u_e(p_1, s_1)] [\bar{u}_\mu(p_2', s_2')\gamma_\mu u_\mu(p_2, s_2)] . \tag{1.4}$$

Removing the overall i gives the amplitude

$$\boxed{\mathcal{M}_{s_1' s_2'; s_1 s_2} = \frac{e^2}{q^2} [\bar{u}_e(p_1', s_1')\gamma^\mu u_e(p_1, s_1)][\bar{u}_\mu(p_2', s_2')\gamma_\mu u_\mu(p_2, s_2)].} \tag{1.5}$$

2 Unpolarized calculation using traces

We first calculate the unpolarized result as a reference. Sum over unobserved final spins and average over the initial spins of the unpolarized incoming particles. Completeness gives

$$\sum_s u_e(p,s)\bar{u}_e(p,s) = \not{p}, \qquad \sum_s u_\mu(p,s)\bar{u}_\mu(p,s) = \not{p} + m. \tag{2.1}$$

There are two initial spin states for each incoming fermion, hence the average factor 1/4. With $q^4 \equiv (q^2)^2$,

$$\overline{|\mathcal{M}|^2} \equiv \frac{1}{4}\sum_{\text{all spins}} |\mathcal{M}|^2 \tag{2.2}$$

$$= \frac{e^4}{4q^4}\,\mathrm{Tr}[\not{p}_1'\gamma^\mu\ \not{p}_1\gamma^\nu]\,\mathrm{Tr}[(\not{p}_2' + m)\gamma_\mu(\not{p}_2 + m)\gamma_\nu] \tag{2.3}$$

$$= \frac{e^4}{4q^4} L_e^{\mu\nu} L_{\mu\nu}^{(\mu)}. \tag{2.4}$$

The label (μ) on $L^{(\mu)}$ identifies the muon tensor; the repeated Lorentz indices are contracted.

2.1 Evaluate the muon trace

Expand the two factors $(\not{p}_2' + m)$ and $(\not{p}_2 + m)$:

$$L_{\mu\nu}^{(\mu)} = \mathrm{Tr}[\not{p}_2'\gamma_\mu\ \not{p}_2\gamma_\nu] + m\,\mathrm{Tr}[\not{p}_2'\gamma_\mu\gamma_\nu] + m\,\mathrm{Tr}[\gamma_\mu\ \not{p}_2\gamma_\nu] + m^2\,\mathrm{Tr}[\gamma_\mu\gamma_\nu]. \tag{2.5}$$

The middle two terms contain an odd number of gamma matrices, so their traces vanish. Use

$$\mathrm{Tr}(\gamma^\alpha\gamma^\mu\gamma^\beta\gamma^\nu) = 4(g^{\alpha\mu}g^{\beta\nu} - g^{\alpha\beta}g^{\mu\nu} + g^{\alpha\nu}g^{\mu\beta}), \tag{2.6}$$

$$\mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4g^{\mu\nu}. \tag{2.7}$$

Writing $\not{p}_2' = p_{2\alpha}'\gamma^\alpha$ and $\not{p}_2 = p_{2\beta}\gamma^\beta$, the four-gamma term becomes

$$\mathrm{Tr}[\not{p}_2'\gamma^\mu\ \not{p}_2\gamma^\nu] = 4p_{2\alpha}'p_{2\beta}(g^{\alpha\mu}g^{\beta\nu} - g^{\alpha\beta}g^{\mu\nu} + g^{\alpha\nu}g^{\mu\beta}) \tag{2.8}$$

$$= 4(p_2'^{\mu}p_2^{\nu} + p_2'^{\nu}p_2^{\mu} - g^{\mu\nu}p_2'\cdot p_2). \tag{2.9}$$

The remaining two-gamma term is $m^2\,\mathrm{Tr}(\gamma^\mu\gamma^\nu) = 4m^2g^{\mu\nu}$. Combining them gives

$$L^{(\mu)\mu\nu} = 4\left[p_2'^{\mu}p_2^{\nu} + p_2'^{\nu}p_2^{\mu} - g^{\mu\nu}(p_2'\cdot p_2 - m^2)\right]. \tag{2.10}$$

2.2 Evaluate the electron trace

Since $m_e = 0$, the electron trace is

$$L_e^{\mu\nu} = \mathrm{Tr}[\not{p}_1'\gamma^\mu\ \not{p}_1\gamma^\nu] \tag{2.11}$$

$$= p_{1\alpha}'p_{1\beta}\,\mathrm{Tr}[\gamma^\alpha\gamma^\mu\gamma^\beta\gamma^\nu] \tag{2.12}$$

$$= 4\left[p_1'^{\mu}p_1^{\nu} + p_1'^{\nu}p_1^{\mu} - g^{\mu\nu}(p_1'\cdot p_1)\right]. \tag{2.13}$$

2.3 Contract the traces

For clarity, define $a = p_1' \cdot p_1$ and $b = p_2' \cdot p_2$. Substituting Eqs. (2.10) and (2.13) gives

$$L_e^{\mu\nu} L_{\mu\nu}^{(\mu)} = 16[p_1'^{\mu} p_1^{\nu} + p_1'^{\nu} p_1^{\mu} - g^{\mu\nu} a] \tag{2.14}$$

$$\times [p_{2\mu}' p_{2\nu} + p_{2\nu}' p_{2\mu} - g_{\mu\nu}(b - m^2)]. \tag{2.15}$$

Multiply each term in the first bracket by each term in the second. The four momentum-by-momentum products give

$$2(p_1' \cdot p_2')(p_1 \cdot p_2) + 2(p_1' \cdot p_2)(p_1 \cdot p_2'). \tag{2.16}$$

The terms containing a metric give, in order,

$$-2a(b - m^2) - 2ab + 4a(b - m^2) \tag{2.17}$$

$$= -2ab + 2am^2 - 2ab + 4ab - 4am^2 \tag{2.18}$$

$$= -2am^2. \tag{2.19}$$

Therefore

$$L_e^{\mu\nu} L_{\mu\nu}^{(\mu)} = 16 \left[2(p_1' \cdot p_2')(p_1 \cdot p_2) + 2(p_1' \cdot p_2)(p_1 \cdot p_2') - 2m^2(p_1' \cdot p_1) \right] \tag{2.20}$$

$$= 32 \left[(p_1 \cdot p_2')(p_1' \cdot p_2) + (p_1 \cdot p_2)(p_1' \cdot p_2') - m^2(p_1 \cdot p_1') \right]. \tag{2.21}$$

Insert this result into Eq. (2.4):

$$\boxed{\overline{|\mathcal{M}|^2} = \frac{8e^4}{q^4} \left[(p_1 \cdot p_2')(p_1' \cdot p_2) + (p_1 \cdot p_2)(p_1' \cdot p_2') - m^2(p_1 \cdot p_1') \right].} \tag{2.22}$$

This is the invariant result from the original calculation.

2.4 Centre-of-momentum kinematics and final result

In the centre-of-momentum frame, the incoming electron travels along $+z$ and the incoming muon along $-z$. The outgoing electron is scattered through angle θ in the xz plane.

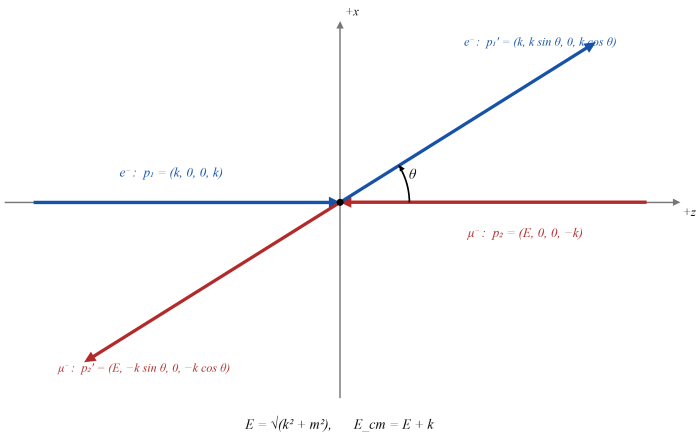


Figure 2: Centre-of-momentum geometry.

The four-momenta are

$$p_1 = (k, 0, 0, k), \qquad p_2 = (E, 0, 0, -k), \tag{2.23}$$

$$p_1' = (k, k\sin\theta, 0, k\cos\theta), \qquad p_2' = (E, -k\sin\theta, 0, -k\cos\theta), \tag{2.24}$$

with $E = \sqrt{k^2 + m^2}$. Using $p\cdot r = p^0 r^0 - \boldsymbol{p}\cdot\boldsymbol{r}$, calculate the scalar products:

$$p_1\cdot p_2 = kE - (0,0,k)\cdot(0,0,-k) = kE + k^2 = k(E+k), \tag{2.25}$$

$$p_1'\cdot p_2' = kE - (k\sin\theta, 0, k\cos\theta)\cdot(-k\sin\theta, 0, -k\cos\theta) \tag{2.26}$$

$$= kE + k^2(\sin^2\theta + \cos^2\theta) = k(E+k), \tag{2.27}$$

$$p_1\cdot p_2' = kE - (0,0,k)\cdot(-k\sin\theta, 0, -k\cos\theta) = k(E + k\cos\theta), \tag{2.28}$$

$$p_1'\cdot p_2 = kE - (k\sin\theta, 0, k\cos\theta)\cdot(0,0,-k) = k(E + k\cos\theta), \tag{2.29}$$

$$p_1\cdot p_1' = k^2 - (0,0,k)\cdot(k\sin\theta, 0, k\cos\theta) = k^2(1-\cos\theta). \tag{2.30}$$

The momentum transfer is

$$q = p_1' - p_1 \tag{2.31}$$

$$= (k, k\sin\theta, 0, k\cos\theta) - (k,0,0,k) \tag{2.32}$$

$$= (0, k\sin\theta, 0, k(\cos\theta - 1)), \tag{2.33}$$

$$q^2 = -k^2\sin^2\theta - k^2(\cos\theta - 1)^2 \tag{2.34}$$

$$= -k^2[\sin^2\theta + \cos^2\theta - 2\cos\theta + 1] \tag{2.35}$$

$$= -2k^2(1-\cos\theta), \tag{2.36}$$

$$q^4 = (q^2)^2 = 4k^4(1-\cos\theta)^2. \tag{2.37}$$

Substitute the scalar products into Eq. (2.22):

$$\overline{|\mathcal{M}|^2} = \frac{8e^4}{q^4}\left[k^2(E + k\cos\theta)^2 + k^2(E+k)^2 - m^2k^2(1-\cos\theta)\right] \tag{2.38}$$

$$= \frac{8e^4k^2}{4k^4(1-\cos\theta)^2}\left[(E + k\cos\theta)^2 + (E+k)^2 - m^2(1-\cos\theta)\right]. \tag{2.39}$$

Thus the final spin-averaged result is

$$\boxed{\overline{|\mathcal{M}|^2} = \frac{2e^4}{k^2(1-\cos\theta)^2}\left[(E+k)^2 + (E + k\cos\theta)^2 - m^2(1-\cos\theta)\right].} \tag{2.40}$$

3 Helicity amplitudes

We now calculate individual helicity amplitudes and check their spin average against the unpolarized result in Eq. (2.40).

3.1 Helicity spinors and centre-of-momentum kinematics

We label each fermion by its helicity $\lambda = +$ or $-$, corresponding to $h = +1$ or $h = -1$ in formulas that use h; the physical helicity eigenvalue is $h/2$. For a momentum direction (ϑ, φ),

define the two-component helicity spinors

$$\xi_+(p) = \begin{pmatrix} \cos(\vartheta/2) \\ e^{\mathrm{i}\varphi}\sin(\vartheta/2) \end{pmatrix}, \qquad \xi_-(p) = \begin{pmatrix} -e^{-\mathrm{i}\varphi}\sin(\vartheta/2) \\ \cos(\vartheta/2) \end{pmatrix}. \tag{3.1}$$

They represent positive and negative helicity, respectively. Their angular dependence specifies how the spinor components change as the momentum direction changes.

In the Weyl representation, positive-energy fermion spinors are

$$u_+(p) = \begin{pmatrix} \sqrt{E_f - |\boldsymbol{p}|}\,\xi_+ \\ \sqrt{E_f + |\boldsymbol{p}|}\,\xi_+ \end{pmatrix}, \qquad u_-(p) = \begin{pmatrix} \sqrt{E_f + |\boldsymbol{p}|}\,\xi_- \\ \sqrt{E_f - |\boldsymbol{p}|}\,\xi_- \end{pmatrix}. \tag{3.2}$$

For the electron, $E_f = |\boldsymbol{p}| = k$, so one Weyl component vanishes for each helicity. For the muon, $E_f = E = \sqrt{k^2 + m^2}$ and $|\boldsymbol{p}| = k$, so both components remain. Thus the massless-electron approximation and the nonzero muon mass enter through the spinor components here.

We use the centre-of-momentum geometry introduced in Sec. 2.4 and shown in Fig. 2. The four-momenta, including their time components, are

$$p_1 = (k, 0, 0, k), \qquad p_2 = (E, 0, 0, -k), \tag{3.3}$$

$$p_1' = (k, k\sin\theta, 0, k\cos\theta), \qquad p_2' = (E, -k\sin\theta, 0, -k\cos\theta). \tag{3.4}$$

We choose the polar and azimuthal angles to be

$$(\vartheta_{p_1}, \varphi_{p_1}) = (0, 0), \qquad (\vartheta_{p_2}, \varphi_{p_2}) = (\pi, \pi), \tag{3.5}$$

$$(\vartheta_{p_1'}, \varphi_{p_1'}) = (\theta, 0), \qquad (\vartheta_{p_2'}, \varphi_{p_2'}) = (\pi - \theta, \pi). \tag{3.6}$$

At a pole, the azimuth is a phase convention for the spinor rather than a physical feature of the momentum direction. Keeping this convention fixed makes the relative phases between spin states consistent.

3.2 Helicity amplitude and calculation strategy

Replace the spin labels in Eq. (1.5) by $s_1 = \lambda_1$, $s_2 = \lambda_2$, $s_1' = \lambda_3$, and $s_2' = \lambda_4$. The amplitude with all four helicity labels explicit is

$$\boxed{\mathcal{M}_{\lambda_3\lambda_4;\lambda_1\lambda_2} = \frac{e^2}{q^2}[\bar{u}_{e,\lambda_3}(p_1')\gamma^\mu u_{e,\lambda_1}(p_1)][\bar{u}_{\mu,\lambda_4}(p_2')\gamma_\mu u_{\mu,\lambda_2}(p_2)].} \tag{3.7}$$

For a fixed helicity assignment, evaluate it in this order:

1. Choose the four helicities.
2. Substitute the corresponding external spinors from Eq. (3.2).
3. Evaluate the electron and muon currents.

4. Contract their Lorentz indices to obtain the amplitude.

5. Take the absolute square.

Each individual helicity amplitude is calculated using explicit spinors. The completeness relation $\sum_s u(p,s)\bar{u}(p,s) = \not{p} + m$ was used in Sec. 2 to sum over unobserved spins and average over unpolarized initial spins. Keeping the helicity amplitudes separate also allows coherent superpositions of different helicities to interfere, as required for transverse-spin states.

3.3 Electron current

Set $c = \cos(\theta/2)$ and $s = \sin(\theta/2)$. The incoming electron moves along $+z$, giving

$$\xi_+(p_1) = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad \xi_-(p_1) = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \tag{3.8}$$

The outgoing electron has direction angles $(\theta, 0)$, so

$$\xi_+(p_1') = \begin{pmatrix} c \\ s \end{pmatrix}, \qquad \xi_-(p_1') = \begin{pmatrix} -s \\ c \end{pmatrix}. \tag{3.9}$$

Their Hermitian conjugates are

$$\xi_+^\dagger(p_1') = (c, s), \qquad \xi_-^\dagger(p_1') = (-s, c). \tag{3.10}$$

The Pauli matrices needed for the spatial current components are

$$\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma^2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{3.11}$$

For each incoming and outgoing helicity, define the two-component products

$$X^0 = \xi_{h'}^\dagger(p_1')\xi_h(p_1), \qquad X^i = \xi_{h'}^\dagger(p_1')\sigma^i\xi_h(p_1) \quad (i = 1, 2, 3). \tag{3.12}$$

As an example, for the ++ assignment,

$$X^0 = (c, s)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = c, \qquad X^1 = (c, s)\begin{pmatrix} 0 \\ 1 \end{pmatrix} = s, \tag{3.13}$$

$$X^2 = (c, s)\begin{pmatrix} 0 \\ \mathrm{i} \end{pmatrix} = \mathrm{i}s, \qquad X^3 = (c, s)\begin{pmatrix} 1 \\ 0 \end{pmatrix} = c. \tag{3.14}$$

For a massless positive-helicity electron, the nonzero Weyl component gives $\bar{u}\gamma^\mu u = 2k(X^0, X^1, X^2, X^3)$, hence

$$J_e^\mu(p_1'+, p_1+) = (2kc, 2ks, 2\mathrm{i}ks, 2kc). \tag{3.15}$$

For clarity, here is a second explicit example, with outgoing helicity $-$ and incoming helicity $+$. In this case $\xi_-^\dagger(p_1') = (-s, c)$ and $\xi_+(p_1) = (1, 0)^T$, so

$$X^0 = (-s, c)\begin{pmatrix}1\\0\end{pmatrix} = -s, \qquad X^1 = (-s, c)\begin{pmatrix}0\\1\end{pmatrix} = c, \tag{3.16}$$

$$X^2 = (-s, c)\begin{pmatrix}0\\\mathrm{i}\end{pmatrix} = \mathrm{i}c, \qquad X^3 = (-s, c)\begin{pmatrix}1\\0\end{pmatrix} = -s. \tag{3.17}$$

Thus $(X^0, X^1, X^2, X^3)_{-+} = (-s, c, \mathrm{i}c, -s)$. Repeating the same multiplication for the other assignments gives

$$(X^0, X^1, X^2, X^3)_{++} = (c, s, \mathrm{i}s, c), \qquad (X^0, X^1, X^2, X^3)_{--} = (c, -s, \mathrm{i}s, -c), \tag{3.18}$$

$$(X^0, X^1, X^2, X^3)_{+-} = (s, c, -\mathrm{i}c, -s), \qquad (X^0, X^1, X^2, X^3)_{-+} = (-s, c, \mathrm{i}c, -s). \tag{3.19}$$

The first sign labels the outgoing helicity and the second the incoming helicity.

From the Pauli products to the full electron currents. Define $J^\mu_{e,h'h} = \bar{u}_{h'}(p_1')\gamma^\mu u_h(p_1)$, with the outgoing helicity first. In the massless limit, Eq. (3.2) becomes

$$u_+(p) = \sqrt{2k}\begin{pmatrix}0_2\\\xi_+(p)\end{pmatrix}, \qquad u_-(p) = \sqrt{2k}\begin{pmatrix}\xi_-(p)\\0_2\end{pmatrix}, \qquad 0_2 = \begin{pmatrix}0\\0\end{pmatrix}. \tag{3.20}$$

The matrices entering the current are block diagonal:

$$\gamma^0\gamma^0 = \begin{pmatrix}I_2 & 0\\0 & I_2\end{pmatrix}, \qquad \gamma^0\gamma^i = \begin{pmatrix}-\sigma^i & 0\\0 & \sigma^i\end{pmatrix}, \qquad i = 1, 2, 3. \tag{3.21}$$

Since $J^\mu = u'^\dagger\gamma^0\gamma^\mu u$, the lower Weyl block gives $J^0_{e,++} = 2kX^0_{++}$ and $J^i_{e,++} = 2kX^i_{++}$. The upper block gives $J^0_{e,--} = 2kX^0_{--}$ and $J^i_{e,--} = -2kX^i_{--}$: only the spatial components acquire a minus sign. Substitution therefore gives

$$\boxed{\begin{aligned} J^\mu_{e,++} &= 2k(X^0_{++}, X^1_{++}, X^2_{++}, X^3_{++})\\ &= (2kc, 2ks, 2\mathrm{i}ks, 2kc),\\ J^\mu_{e,--} &= 2k(X^0_{--}, -X^1_{--}, -X^2_{--}, -X^3_{--})\\ &= (2kc, 2ks, -2\mathrm{i}ks, 2kc). \end{aligned}} \tag{3.22}$$

For a helicity-flip transition, the incoming and outgoing spinors occupy opposite Weyl blocks. Every $\gamma^0\gamma^\mu$ above is block diagonal, so it cannot connect those blocks. Explicitly, writing $D_i = \mathrm{diag}(-\sigma^i, \sigma^i)$,

$$J^0_{e,+-} = 2k(0_2^\dagger, \xi_+'^\dagger)\begin{pmatrix}\xi_-\\0_2\end{pmatrix} = 0, \qquad J^i_{e,+-} = 2k(0_2^\dagger, \xi_+'^\dagger)D_i\begin{pmatrix}\xi_-\\0_2\end{pmatrix} = 0, \tag{3.23}$$

$$J^0_{e,-+} = 2k(\xi_-'^\dagger, 0_2^\dagger)\begin{pmatrix}0_2\\\xi_+\end{pmatrix} = 0, \qquad J^i_{e,-+} = 2k(\xi_-'^\dagger, 0_2^\dagger)D_i\begin{pmatrix}0_2\\\xi_+\end{pmatrix} = 0. \tag{3.24}$$

$$\boxed{J^\mu_{e,+-} = 0, \qquad J^\mu_{e,-+} = 0.} \tag{3.25}$$

Here $\xi_h' = \xi_h(p_1')$ and $\xi_h = \xi_h(p_1)$. This is the explicit spinor form of helicity conservation at the massless electron vector vertex. The Pauli products X^μ_{+-} and X^μ_{-+} can be nonzero, while the complete currents vanish because of their Weyl-block structure.

3.4 Muon current

For the muon, both Weyl components contribute.

Write $u_h = (a_h\xi_h, b_h\xi_h)^T$, where $a_h = \sqrt{E - hk}$ and $b_h = \sqrt{E + hk}$ for $h = +1, -1$. Multiplying the upper and lower two-component blocks gives

$$\bar{u}_{h'}(p_2')\gamma^0 u_h(p_2) = (b_{h'}b_h + a_{h'}a_h)\,\xi^\dagger_{h'}(p_2')\xi_h(p_2), \tag{3.26}$$

$$\bar{u}_{h'}(p_2')\gamma^i u_h(p_2) = (b_{h'}b_h - a_{h'}a_h)\,\xi^\dagger_{h'}(p_2')\sigma^i\xi_h(p_2), \qquad i = 1, 2, 3. \tag{3.27}$$

With $a_\pm = \sqrt{E \mp k}$ and $b_\pm = \sqrt{E \pm k}$, the coefficients multiplying the time and spatial two-component products are

$$A_{++} = A_{--} = b_+^2 + a_+^2 = 2E, \tag{3.28}$$

$$B_{++} = b_+^2 - a_+^2 = 2k, \qquad B_{--} = b_-^2 - a_-^2 = -2k, \tag{3.29}$$

$$A_{+-} = A_{-+} = b_+b_- + a_+a_- = 2m, \qquad B_{+-} = B_{-+} = b_+b_- - a_+a_- = 0. \tag{3.30}$$

The resulting four muon currents are

$$\boxed{\begin{array}{ll} J^\mu_{\mu,++} = (2Ec, -2ks, 2iks, -2kc), & J^\mu_{\mu,--} = (2Ec, -2ks, -2iks, -2kc), \\ J^\mu_{\mu,+-} = (2ms, 0, 0, 0), & J^\mu_{\mu,-+} = (-2ms, 0, 0, 0). \end{array}} \tag{3.31}$$

Here the first helicity is outgoing and the second is incoming. In the present centre-of-momentum kinematics, **the muon helicity-flip currents have only a time component and are proportional to the mass** m. They therefore vanish in the limit $m \to 0$.

- The $++$ current at $\theta = 0$ reduces to $(2E, 0, 0, -2k)$, as expected for a muon moving along $-z$.

3.5 The helicity-amplitude table for this process

The amplitude is the product of the electron and muon currents,

$$\mathcal{M}_{\lambda_3\lambda_4;\lambda_1\lambda_2} = \frac{e^2}{q^2} J_e \cdot J_\mu \tag{3.32}$$

For example, for initial helicities $(\lambda_1, \lambda_2) = (+, +)$ and final helicities $(\lambda_3, \lambda_4) = (+, +)$,

$$J_e \cdot J_\mu = J_e^0 J_\mu^0 - J_e^1 J_\mu^1 - J_e^2 J_\mu^2 - J_e^3 J_\mu^3 \tag{3.33}$$

$$= (2kc)(2Ec) - (2ks)(-2ks) - (2iks)(2iks) - (2kc)(-2kc) \tag{3.34}$$

$$= 4kEc^2 + 4k^2s^2 + 4k^2s^2 + 4k^2c^2 \tag{3.35}$$

$$= 4k\left[(E + k)c^2 + 2ks^2\right]. \tag{3.36}$$

Define

$$F = (E + k)c^2 + 2ks^2, \qquad G = (E + k)c^2, \qquad H = m\,s\,c. \tag{3.37}$$

The same four multiplications for each allowed helicity choice give the table below. Rows are the incoming pair (λ_1, λ_2) and columns are the outgoing pair (λ_3, λ_4). The entries are $q^2\mathcal{M}/(4ke^2)$; multiply each by $4ke^2/q^2$ to recover the corresponding amplitude.

$(\lambda_1,\lambda_2)\backslash(\lambda_3,\lambda_4)$	$++$	$+-$	$-+$	$--$
$++$	F	$-H$	0	0
$+-$	H	G	0	0
$-+$	0	0	G	$-H$
$--$	0	0	H	F

(3.38)

The table contains

- two entries equal to F;
- two entries equal to G;
- four entries with magnitude H;
- zeros in all remaining positions.

Different final helicity states are orthogonal. When the final spins are unobserved, we therefore **sum their probabilities**. For unpolarized incoming particles, we also average over the initial helicity states. The spin average is defined by

$$\overline{|\mathcal{M}|^2}_{\text{helicity}} = \frac{1}{4}\sum_{\lambda_1,\lambda_2}\sum_{\lambda_3,\lambda_4}|\mathcal{M}_{\lambda_3\lambda_4;\lambda_1\lambda_2}|^2\,. \tag{3.39}$$

Every amplitude in the table has the common factor $4ke^2/q^2$. For each fixed incoming pair, sum the squared magnitudes of the four entries across its row. Since F, G, and H are real, these row sums are

$$(\lambda_1,\lambda_2) = (+,+): \quad F^2 + (-H)^2 + 0 + 0 = F^2 + H^2, \tag{3.40}$$
$$(\lambda_1,\lambda_2) = (+,-): \quad H^2 + G^2 + 0 + 0 = H^2 + G^2, \tag{3.41}$$
$$(\lambda_1,\lambda_2) = (-,+): \quad 0 + 0 + G^2 + (-H)^2 = G^2 + H^2, \tag{3.42}$$
$$(\lambda_1,\lambda_2) = (-,-): \quad 0 + 0 + H^2 + F^2 = H^2 + F^2. \tag{3.43}$$

$$\overline{|\mathcal{M}|^2}_{\text{helicity}} = \frac{1}{4}\left(\frac{4ke^2}{q^2}\right)^2 (2F^2 + 2G^2 + 4H^2) \tag{3.44}$$
$$= \frac{8k^2e^4}{q^4}(F^2 + G^2 + 2H^2). \tag{3.45}$$

The factor $1/4$ arises because the incoming electron and muon each have two spin states, giving four equally weighted initial helicity pairs.

Since $c^2 = 1 - s^2$ and $\cos\theta = 1 - 2s^2$, direct expansion gives

$$F^2 + G^2 + 2H^2 = (E + k - 2ks^2)^2 + (E + k)^2 - 2m^2 s^2 \tag{3.46}$$

$$= (E + k\cos\theta)^2 + (E + k)^2 - m^2(1 - \cos\theta). \tag{3.47}$$

Substituting this expression into the helicity spin average and using $q^4 = 4k^4(1 - \cos\theta)^2$ from Eq. (2.37) gives the final result

$$\boxed{\overline{|\mathcal{M}|^2}_{\text{helicity}} = \frac{2e^4}{k^2(1 - \cos\theta)^2} \quad \times \left[(E + k)^2 + (E + k\cos\theta)^2 - m^2(1 - \cos\theta)\right].} \tag{3.48}$$

This agrees with the trace result in Eq. (2.40), checking the unpolarized normalization and the sum of squared helicity amplitudes.

4 Transverse-spin states and interference

We now use the helicity amplitudes to describe a muon whose spin points perpendicular to its momentum. The key rule is to combine amplitudes first and take the absolute square afterwards.

4.1 A transverse state contains both helicities

A general normalized spin state is

$$|\psi\rangle = \alpha|+\rangle + \beta|-\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1. \tag{4.1}$$

Here $|\alpha|^2$ and $|\beta|^2$ are the two helicity probabilities. The corresponding external spinor is $u_\psi = \alpha u_+ + \beta u_-$. For a pure transverse state, the two probabilities are equal. Their relative phase determines the transverse direction.

For our incoming muon, the chosen phases give $\xi_+(p_2) = (0, -1)^T$ and $\xi_-(p_2) = (1, 0)^T$. Their difference is proportional to $(1, 1)^T$, the spin state along $+x$. Thus the physical x states in this convention are

$$u_{\mu,+x} = \frac{u_+ - u_-}{\sqrt{2}}, \qquad u_{\mu,-x} = \frac{u_+ + u_-}{\sqrt{2}}. \tag{4.2}$$

All spinors here have momentum p_2. The minus sign for $+x$ follows from this phase convention.

4.2 Adding amplitudes and finding interference

Fix the electron's incoming helicity and the final helicities. Let A_+ and A_- be the amplitudes for incoming muon helicities $+$ and $-$. Linearity gives

$$\mathcal{M}_\psi = \alpha A_+ + \beta A_-. \tag{4.3}$$

For the $+x$ state, this becomes

$$\mathcal{M}_{+x} = \frac{A_+ - A_-}{\sqrt{2}}. \tag{4.4}$$

For example, with incoming electron helicity + and final state ++, the table gives $A_+ = CF$ and $A_- = CH$, where $C = 4ke^2/q^2$. Therefore,

$$\mathcal{M}_{++;+,+x} = \frac{C}{\sqrt{2}}(F - H). \tag{4.5}$$

Taking the absolute square gives

$$|\mathcal{M}_{+x}|^2 = \frac{1}{2}\left(|A_+|^2 + |A_-|^2 - 2\,\mathrm{Re}[A_+ A_-^*]\right). \tag{4.6}$$

The last term is interference between two helicities leading to the same final state. An unpolarized muon gives $(|A_+|^2 + |A_-|^2)/2$, without this interference term.

When final spins are unobserved, sum their probabilities. For incoming electron helicities + and −, respectively, the table gives

$$P_+ = \frac{C^2}{2}\left[(F - H)^2 + (G + H)^2\right], \tag{4.7}$$

$$P_- = \frac{C^2}{2}\left[(G - H)^2 + (F + H)^2\right]. \tag{4.8}$$

For an unpolarized electron, average these two expressions:

$$\frac{P_+ + P_-}{2} = \frac{C^2}{2}(F^2 + G^2 + 2H^2) = \overline{|\mathcal{M}|^2}_{\text{unpolarized}}. \tag{4.9}$$

The interference contributions cancel. In our tree-level calculation, polarizing only the muon transversely therefore leaves this spin-summed result unchanged when the electron is unpolarized.

4.3 Where the density matrix comes from

A density matrix records spin probabilities and their interference. In the ordered helicity basis $(|+\rangle, |-\rangle)$, the state is the coefficient column $(\alpha, \beta)^T$. For a pure state, define

$$\rho = |\psi\rangle\langle\psi| = \begin{pmatrix}\alpha\\ \beta\end{pmatrix}(\alpha^*, \beta^*) = \begin{pmatrix}|\alpha|^2 & \alpha\beta^*\\ \alpha^*\beta & |\beta|^2\end{pmatrix}. \tag{4.10}$$

The diagonal entries are the two helicity probabilities. The off-diagonal entries contain the relative phase.

To write these entries using real numbers, define

$$s_z = |\alpha|^2 - |\beta|^2, \qquad s_x = 2\,\mathrm{Re}(\alpha^*\beta), \qquad s_y = 2\,\mathrm{Im}(\alpha^*\beta). \tag{4.11}$$

Normalization then gives

$$|\alpha|^2 = \frac{1 + s_z}{2}, \qquad |\beta|^2 = \frac{1 - s_z}{2}, \qquad \alpha\beta^* = \frac{s_x - \mathrm{i}s_y}{2}. \tag{4.12}$$

Substitution gives the density matrix

$$\boxed{\rho = \frac{1}{2}\begin{pmatrix} 1+s_z & s_x - \mathrm{i}s_y \\ s_x + \mathrm{i}s_y & 1-s_z \end{pmatrix}.} \tag{4.13}$$

The parameters s_x, s_y, s_z describe polarization in the helicity basis; they are distinct from $s = \sin(\theta/2)$. For our phase convention, the global $+x$ state has $(\alpha, \beta) = (1, -1)/\sqrt{2}$, so $s_x = -1$. For comparison,

$$\rho_{+x} = \frac{1}{2}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \rho_{\text{unpolarized}} = \frac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}. \tag{4.14}$$

Both have equal helicity probabilities. Only the transverse pure state has off-diagonal entries.

For a beam containing different spin states, average their density matrices with their probabilities. The same matrix form applies. A fully polarized state has $|\boldsymbol{s}| = 1$, a partially polarized beam has $0 < |\boldsymbol{s}| < 1$, and an unpolarized beam has $\boldsymbol{s} = 0$.

For fixed electron helicity and final state, the squared amplitude obtained with this matrix is

$$\begin{aligned} \mathcal{P}_f(\rho) = \rho_{++}|A_+|^2 + \rho_{--}|A_-|^2 \\ + \rho_{+-}A_+A_-^* + \rho_{-+}A_-A_+^*. \end{aligned} \tag{4.15}$$

The first two terms weight the helicity probabilities. The last two terms give interference. Inserting ρ_{+x} reproduces $|A_+ - A_-|^2/2$; inserting $\rho_{\text{unpolarized}}$ gives $(|A_+|^2 + |A_-|^2)/2$.